

Lecture outline

- a. List and describe the four types of hypersensitivity reactions.
- b. Define the terms hypersensitivity, allergy, and allergen.
- d. Describe the roles of IgE, cross-linking, and mast cells in type I hypersensitivity.
- e. Describe the roles of antibody and complement in type II hypersensitivity.
- f. Describe the roles of immune complexes, complement, and phagocytes in type III hypersensitivity.
- g. Diagnose each form of hypersensitivity from clinical manifestations.
- h. Give an example of a clinical condition that is characteristic of each form of hypersensitivity.
- i. Describe clinical and laboratory methods used to detect and/or evaluate hypersensitivity.
- j. Describe prophylactic and therapeutic approaches for each type of hypersensitivity.

Types of Hypersensitivity

*hypersensitivity means more immune response in an inappropriate way (allergy)

- **Type I Hypersensitivity:** associated with mast cells, histamine and IgE
- **Type II Hypersensitivity:** (cytotoxic H.);
the reaction to penicillin wherein the drug can bind to red blood cells, causing them to be recognized as different; B cell proliferation will take place and antibodies to the drug are produced. IgG and IgM antibodies bind to these antigens to form complexes that activate classical pathway of complement system.
- **Type III Hypersensitivity:** Immune complexes H.
occurs when antigen-antibody complexes that are not adequately cleared by innate immune cells accumulate, giving rise to an inflammatory response and attraction of leukocytes.
- **Type IV Hypersensitivity:** delayed type hypersensitivity as the reaction takes two to three days to develop. Unlike the other types, it is not antibody mediated but rather is a type of **cell-mediated** response.

Terminology

- **Type I Hypersensitivity:** associated with mast cells, histamine and IgE
- **Allergy and allergic:** although there are 4 types of allergy, the word means type I hypersensitivity
- **Atopy and atopic:** is the term for the genetic trait to have a predisposition for localized anaphylaxis
- **Immediate hypersensitivity:** the rxn is very fast in terms of min
- **Anaphylaxis:** systemic allergy (involve many organs)

Events of Hypersensitivity

- The sequence of events can be divided into 3 phases:

Sensitization, activation and effector phases

- * First exposure $\rightarrow$ immunoglobulin (primary i.r.)
second exposure $\rightarrow$ secondary i.r.
- * First exposure to allergen produce primary i.r.
second exposure makes stronger i.r.
- * symptoms appear after second exposure

Sensitization Phase

~~* IgE~~
* IgE has the least
immunoglobulin
conc

- Attachment of IgE to the receptors at the surface of mast and basophils.
 ↑ primary iir
 ↑ * IgE was IgM
 ↑ results to class switching
 * basophils in blood
 * mast cells in tissues
- May persist for weeks
- Can be achieved by serum transfer with IgE to specific Ag (PK test)
- The responsible cells are: mast cells and basophils

* IgE bind to mast cell through receptor ~~receptor~~ ~~on the~~ mast cell

* synthesization → when IgE bind with IgE R on mast cell

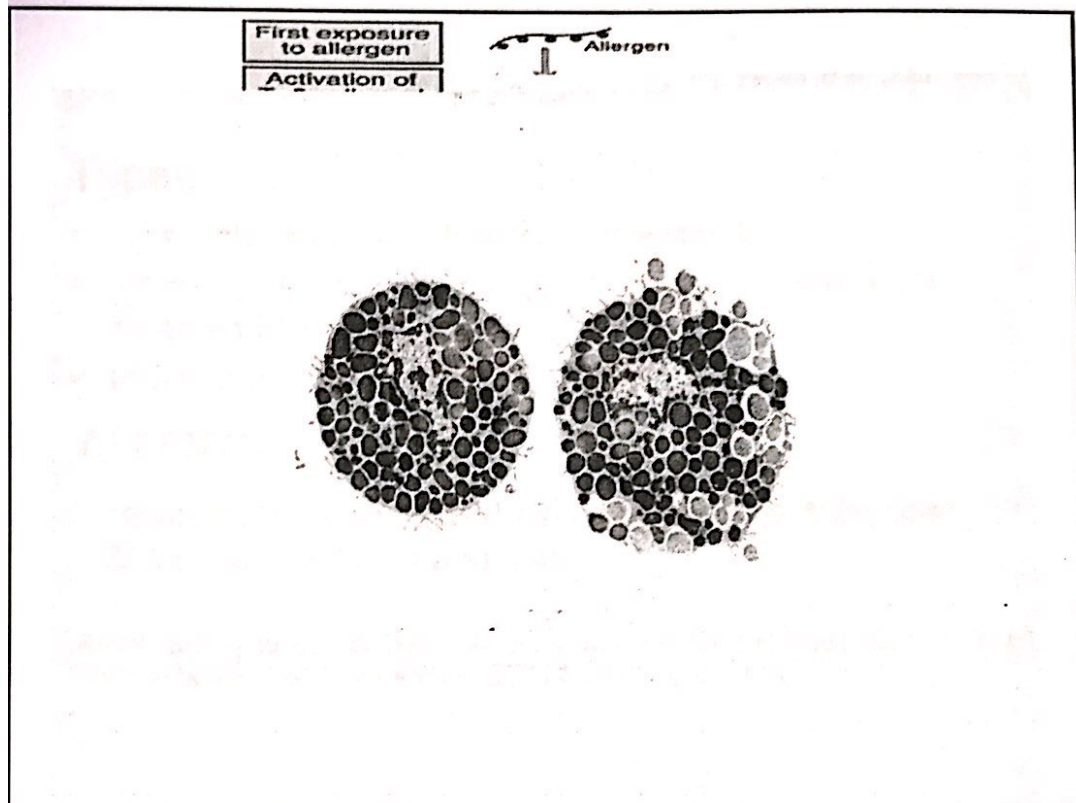
* mast cell is normally filled with granules

Mast cells

- Large mononuclear cells, heavily granulated, stained with basic dyes like toluidine blue
- CTMC: Found around blood vessels in connective tissue with life span about 6 months
- MMC: associated with mucosa lining of gut and lungs with life span about 40 days.
- Originated from bone marrow and mature in the tissues
- Have Fcε R1 with high affinity
- Mature cell can divide in contrast to basophils

meaning that mature basophils don't divide

acidic ←



Basophils

- Originated in bone marrow and mature there
- After maturation, leaves the b. marrow to the blood, cannot divide in contrast to mast cells
- The least count among white blood cells (around 0.5%)
- ^{have IgE R} Resemble mast cells in contents and Have Fcε R1 with high affinity

Allergens

* In the absence of allergens there's no allergy

Types

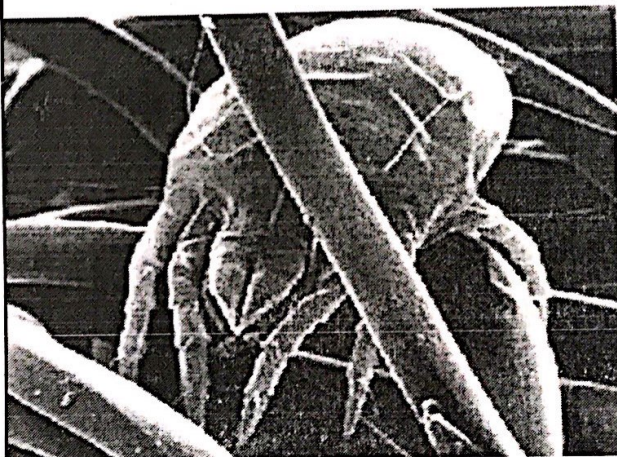
- **injectants** like drugs penicillin for example
- **Inhalants** like mold spores, pollen grains, house dust mites and animal danders
- **Ingestants** like food as strawberry

characteristics

glycoproteins, most of them enzymes, low dose, low M.W, highly soluble and stable

An increasing number of these allergens have now been cloned including Der p1-14 from house dust mites and Lol p1-11 from rye grass pollen.

Allergens



Dermatophagoides
pteronyssinus
(common dust mite)

Example:

Der P1
Fungi

Der P1 is an enzyme allergen from the fecal pellets of the dust mite.

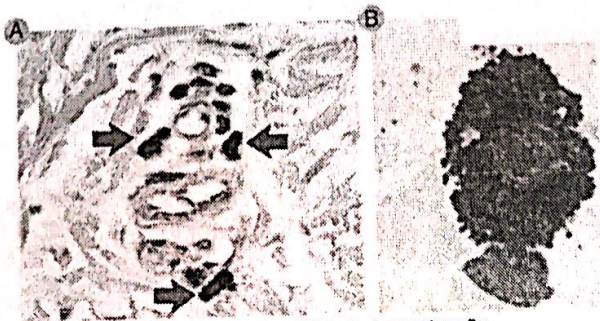
IgE

- The lowest conc. Among all classes of Ig due to inhibition by IFN gamma secreted by Th1
- IgE is Th2 dependent
- Elevated in allergic patients
- Class switch occurs as a result of the release of cytokines IL-4 and IL-13.
 - a. Ab to IL-4 inhibit IgE
 - b. IL-4 gene knockout inhibition of IgE
 - c. The elevated level in allergic patients

* parasite infection is also associated with IgE, so to exclude a parasite infection test is also made with IgE

* most allergic patients have upnormal levels of IgE

Ab IL-4 ~~that~~ that



Abbas et al: Cellular and Molecular Immunology, 7e.
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Genetic Predisposition? why some people have allergy & genetic wise

- **Genetic factors:**

- MHC class II genes, (Immune response genes)
- Gene of IL-4 on chromosome 5 and IL-4R
- FcεR1 gene on chromosome 11
- Treg genes

- **Environmental factors:**

- Children live in cleaner houses tend to have allergy
- Children live in livestock and farms tend to be nonallergic

*all the above are ~~sya~~ sensitizing phase

Activation Phase

Degranulation of mast and basophils

- **IgE involvement:** * IgE is the main source of degranulation

- Polyvalent antigen because at least 2 molecules should cross-link with each other. Monovalent or haptens cannot degranulate mast cells. *Lectin is a protein that recognize sugar
- Lectins cross-link IgE. This explain urticaria induced in some individuals after eating for e.g strawberry. *Lectin is most present in strawberries
- Chemicals that cross-link IgE
- Anti-IgE autoantibodies

- **IgG involvement**

Autoantobodies against Fcε R1 or antiidiotype antibodies

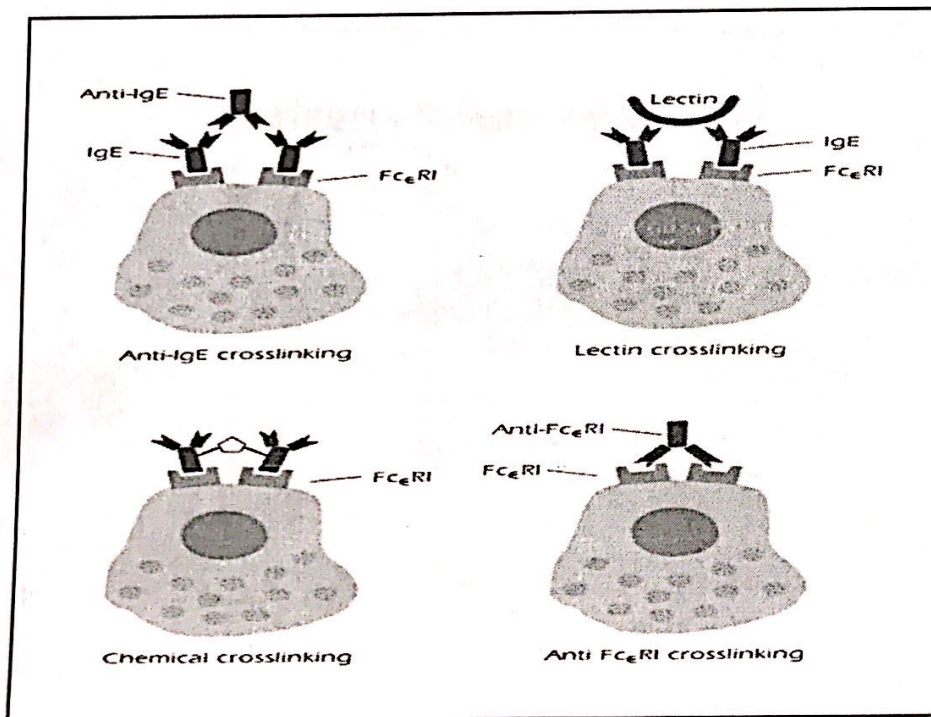
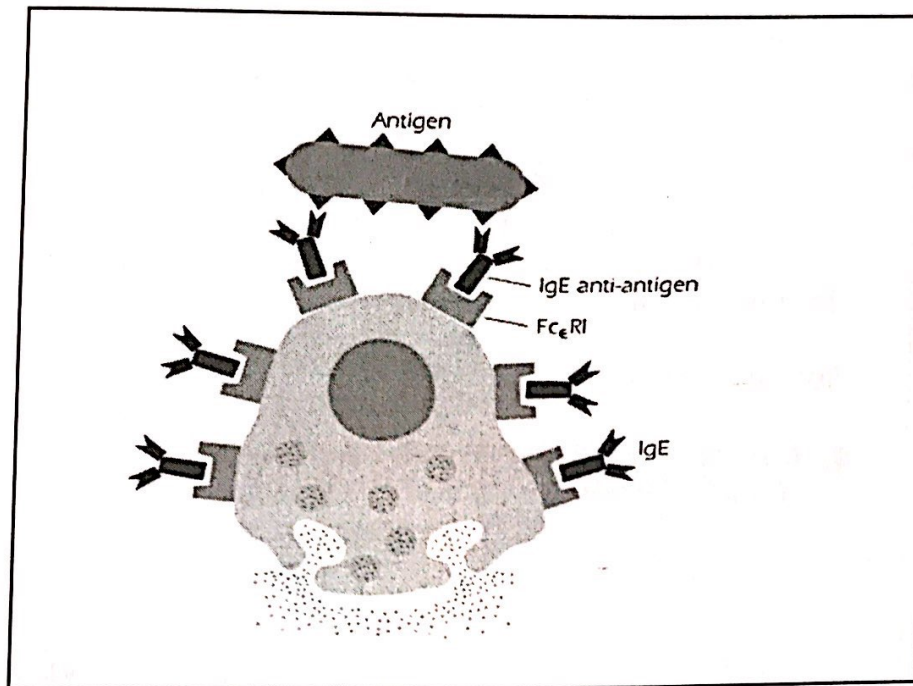
Others:

- Anaphylatoxins bind their specific receptors on mast cells and cause degranulation.
- Various drugs like morphin and codeine bind receptors on mast cells
- Physical factors like cold and heat. Cold-induced urticaria
- Calcium ionophores can activate mast cells and cause rapid influx of calcium ions into the cell leading to degranulation.

mast cell open and release granule ←

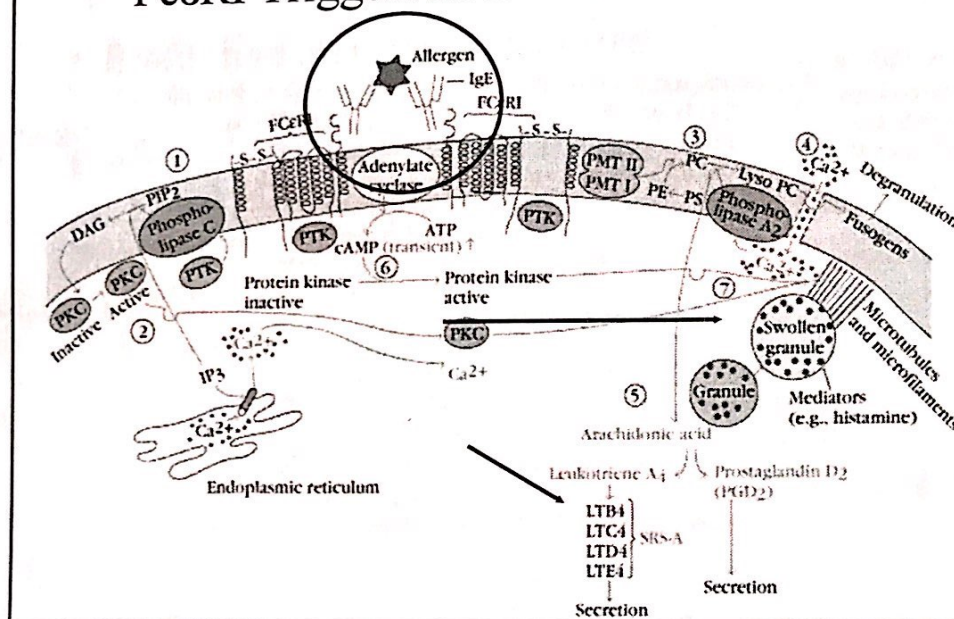
* there's a kind of cockroaches that everyone is sensitive against and it's takes as a positive control

* in a procedure, an allergen is taken from that cockroach and one epitope is taken from it, but there were no allergy against it, which proves that allergy is not monovalent and need more than one epitope to cause allergy



- -methylation of phospholipids
- -rapid influx of Ca^{2+}
- -increase cAMP and decrease in cGMP
temporarily or transient (sustained elevation of cAMP stop degranulation)
- -Movement of granules by microfilaments to cell surface exocytosis
- -Degranulation is a physiological process and no death of the cell occurs but regenerate again.

FcεRI Triggers Release of Mediators



Effector Phase

Activated mast cells secrete a variety of mediators that are responsible for the Symptoms of allergic reactions.

Mediators are two types: preformed and newly formed

Preformed mediators: stored in the granules by electrostatic interaction and comprise the following: Histamine, serotonin, cytokines, proteases and heparin

Newly formed mediators: synthesized upon activation ^{after exposure to allergy} and comprise Prostaglandins, Leukotriens and platelet activating factor

Effector Phase

• Preformed mediators: (granules)

① **Histamine:** derived from Histidine a.a. by decarboxylation

← Stored bound with heparin → amino acid

- Short half life

→ anti coagulation

- Bind H1 receptors on smooth muscles causing

- constriction (asphyxia) and bind endothelial cells causing

increase in V. permeability (drop in B.P.)

- bind H2 receptors involved in mucus secretion and increase in V. permeability.

② **Serotonin:** Present in mast cells of certain species like rodents.

Present in platelets of humans. Its effects like histamine.

③ **Cytokines:** ECF chemotaxis for eosinophils

IL-8 (chemokine) chemotaxis for neutrophils

IL-3, 4, 5 and GM-CSF development of eosinophils and neutrophils, basophils, lymphocytes, Mac, Platelets

TNF-alpha and IL-6 for inflammation.

④ **Heparin:** acidic proteoglycan, bind His and anticoagulant (regulator) (acidic)

⑤ **Proteases:** neutral proteases like trypase and chymase

* not all organs respond to histamine because not all organs have histamine receptors

* H1, H2, H3 are histamine receptors
* smooth muscles exist on veins and urinary bladder

الموجودة في mast cell
preformed

because it's dangerous

* V → vascular
B.P. → blood pressure

serotonin exist in animals but in humans exist in platelets (الصفيحات)

Newly-formed mediators

- Consist of substances synthesized from membrane **arachidonic acid** which is polyunsaturated fatty acid with 4 unsaturated bonds. *exist in membrane and has 4 double bonds*

arachidonic acid make 2 pathways:

- Through cyclooxygenase pathway to give **prostaglandins** and thromboxanes and through lipoxygenase to give **leukotriens**.

90 different forms

the famous use of prostaglandins is in artificial labour (as lipid soluble)

that are associated with allergy

Leukotriens: LTB₄, LTC₄, LTD₄, LTE₄

Metabolites of arachidonic acid-linked some peptides

Cause prolonged constriction of smooth muscle

Responsible for anti-His resistant asthma

Prostaglandins and Thromboxanes: PG and TX specifically PGD₂
Vasoactive causing bronchoconstriction, vasodilation, contraction of smooth muscles, and chemotactic for variety of WBC.

Platelet-activating factor (PAF): Highly potent, derived from phospholipids

cause aggregation of platelets, and release content of His and serotonin,

cause bronchoconstriction and vasodilation

* Leukotriens symptoms are similar to histamine symptoms so when a patient is given anti-His it doesn't work because they have Leukotrien

Late-Phase Reactions

Occurs within 48 h and may persist for several days. Cause inflammation and tissue damage and is caused because of eosinophils and neutrophils

Recruitments of eosinophils due to ECF:

- cause milder wave of s. muscle contraction, produce sustained edema
- bind IgE via FcεRII and bind IgG via FcγR cause activation and degranulation of eosinophils
- Release of LT, PAF, Major basic protein (MBP) that destroy parasites (schistosoma worms) and toxic to respiratory tract, Eosinophil chemotactic factor (ECP) that is neurotoxic and against worms as helminthes
IgE bind the worm---eosinophils bind IgE---MBP and ECP destroy the worm
- Release of Arylsulfatase and histaminase that destroy several mediators

Recruitments of Neutrophils due to IL-8:

- Bind IgG via FcγR
- Release of lysosomal hydrolytic Enzymes
- Release of PAF and LT

Allergic diseases

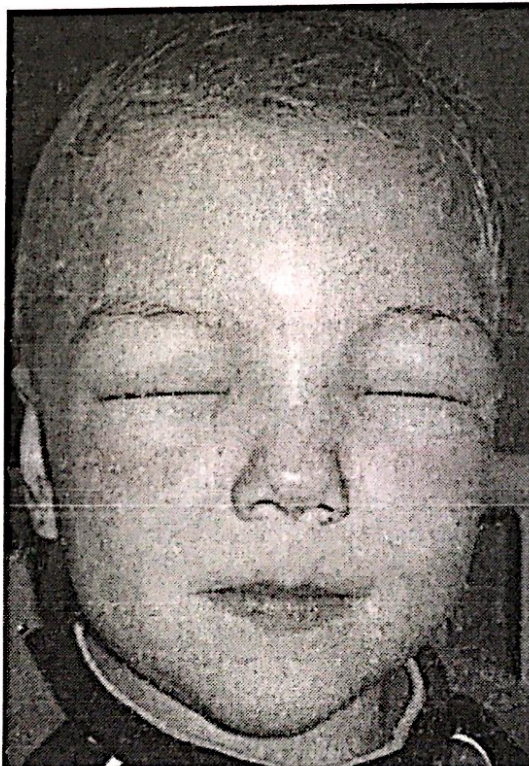
- Localized such as Allergic rhinitis, asthma, atopic dermatitis, food allergy
- Systemic allergy such as anaphylaxis due to drugs
- GIT increase fluid secretion, peristalsis, diarrhea, vomiting
- Lung: decrease in airway, increase in mucus, congestion and blockage
- Blood cells: increase in blood flow, increase V. permeability,, edema, inflammation

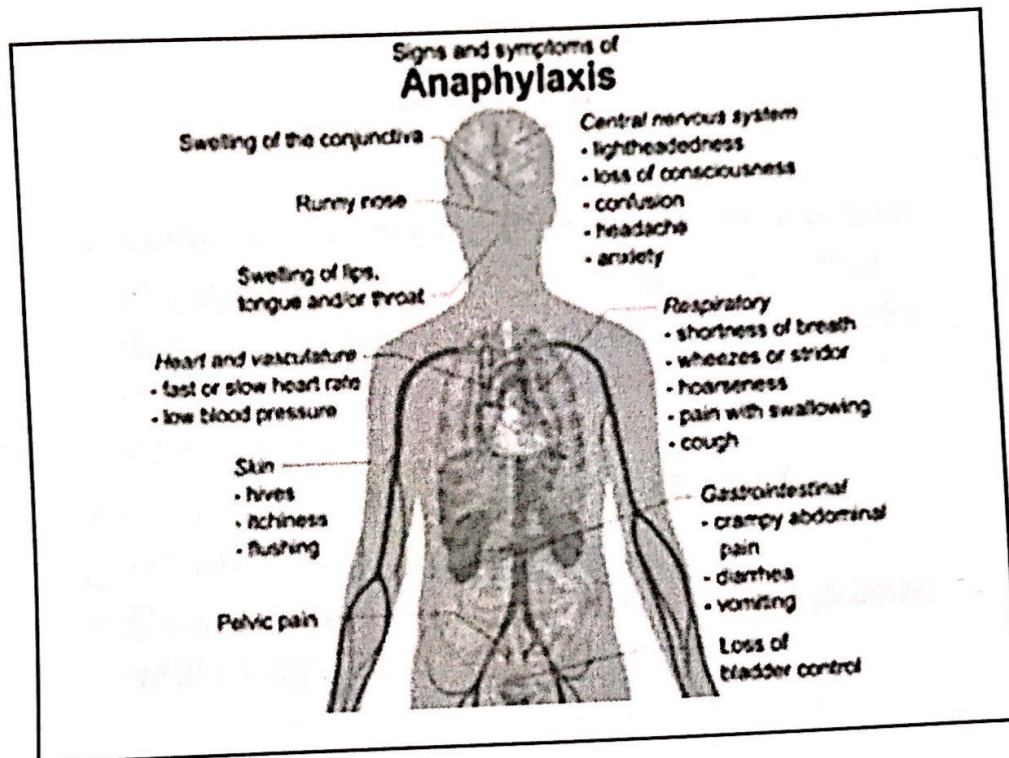
Urticaria & Eczema

- An eruption of itching wheals
- of systemic origin
- It may be due to allergic reaction to:
 - foods
 - drugs
 - foci of infection
 - physical agents (heat, cold, light, friction)
 - psychic stimuli



Angioedema of the face such that the boy cannot open his eyes





Allergic rhinitis

حساسية
الغبار

- commonly known as hay fever
- *Caused by airborne allergens (fungi)
- *Sneezing, coughing, running nose and eyes
- *Antihistamine is the drug of choice:
Benadryl by direct competition to bind H1.
cimetidine by direct competition to bind H2

Asthma

- Asthma is a chronic inflammatory disease of the airways which develops under the allergens influence, manifests with attacks of dyspnea, breathlessness, cough, wheezing, and chest tightness.
- Charac by reversible obstruction of airways causing asphyxia
- Cockroaches calyx, airborne pollen grains, viral antigens, chemicals

the one that everyone is allergic of

Treatment

Pharmacologic Intervention (drugs)

- β -agonists**: affect mast cell —cause sustained $\uparrow$ cAMP —inhibit degranulation
Affect smooth muscles—cause muscle relaxation
Affect b vessels —cause bronchiodilator
Epinephrine in case of emergency and reverse all effects of Histamine $\rightarrow$ **adrenaline**
- Theophyllin and caffeine** inhibit phosphodiesterase that degrade cAMP and prevent degranulation. bronchiodilator
- Membrane stabilizer** like **Na cromolyn** stabilize mast cell membranes
- Corticosteroids**: Potent anti-infl. Drugs + immunosuppressive activity
- LT inhibitors**
- antiHis** is not useful

* caffeine degrade cyclic AMP

كورتيزون

50% of cases
are healed

Immunologic Intervention

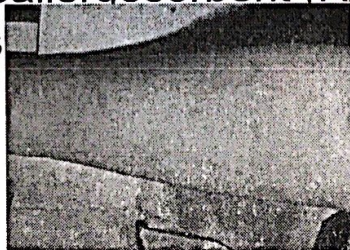
- ← **Hyposensitization (Desensitization)**: small doses, repetitive injections, SC, IgG instead of IgE, Th2 to Th1 *→ reverse of class switch*
- **Humanized anti-IgE Mab** that does not cross link IgE : (Fab) only.
- **Chemically altered allergen**: ragweed pollen denatured by urea: induction of Treg
- **miRNA**: ssRNA, silence gene expression post-transcription

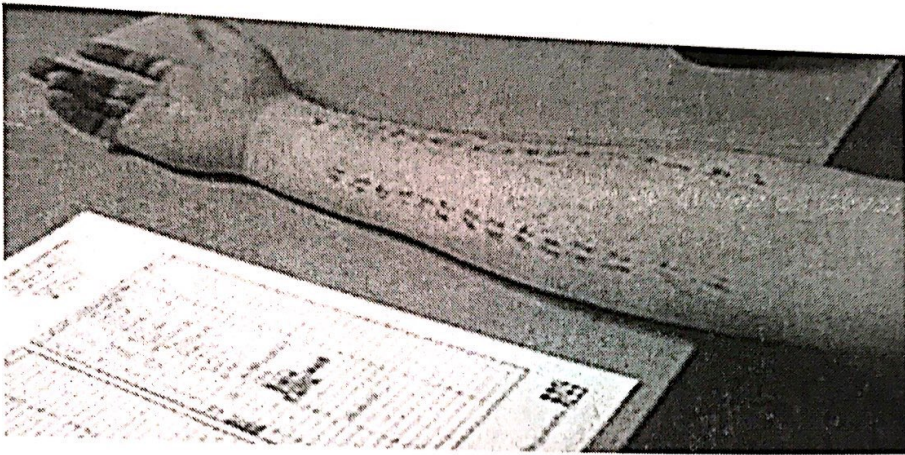
Environmental Intervention:

Avoid exposure to known allergens by using **masks and air filters** or to go to allergen-free area

Diagnosis

- **Skin prick test wheal and flare test**
erythema, edema
- **Radioallergen sorbent (RAST)**: *measurement of IgE*
meas





Beneficial role of IgE

- -Defend external surfaces
- -Against worms: IgG and IgE bind worms ---- attract eosinophils----- eosinophils bind the worm----release of MBP---death of worm

